Latent Arithmetic Measures the Origin: A No-Retraining Intervention on Audio Autoencoders

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***Abstract*—Audio latents are increasingly evaluated by arithmetic: interpolate two codes and decode a mix, subtract a stem's code and decode the residual. We show that one of these two probes is not a property of the autoencoder at all. Shifting the latent origin by the encoded silence $f(0)$, a reparameterization that leaves the autoencoding function and every affine combination with unit coefficient sum exactly unchanged for arbitrary nonlinear encoders and decoders, changes stem-subtraction scores by several decibels. Applying it to five fixed checkpoints, with no retraining, removes the entire apparent advantage of mixing supervision on subtraction: an advantage of $+2.3$ to $+4.0\,$dB becomes -0.07 to $+0.01\,$dB, a reduction on all 49 test tracks, and every model lands within $0.5\,$dB of its own reconstruction of the residual. On convex mixing, which the same reparameterization provably cannot touch, the supervision does help: ground-truth equivariance improves by $+1.7$ to $+2.2\,$dB at two compression rates. The two probes therefore support different conclusions about the same checkpoints, and latent arithmetic should be reported against a coordinate-invariant reference.**

***Index Terms*—representation learning, equivariant representations, audio autoencoders, latent arithmetic, evaluation methodology, source separation**

I. Introduction

Audio mixing is linear on waveforms: $\mathrm{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$. This makes latent arithmetic an attractive interface to a learned audio representation. For an encoder f and decoder g, a latent space is *mixing-equivariant* when

$$g(\alpha f(x_1) + (1-\alpha) f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2, \qquad (1)$$

and a natural companion probe is stem removal by subtraction, $g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}})) \approx x_{\mathrm{res}}$. Where these hold, editing and composition happen in the latent domain without a dedicated separation or synthesis model, a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and in quantized codecs (DAC [3], EnCodec [4]) an interpolated point is not a codebook entry, so code-domain mixing is undefined without leaving the quantizer. Torres et al. [5] induce the property in a consistency autoencoder by decoding averaged latents against the true mix.

This paper asks a prior question: *what do these arithmetic scores measure?* The two probes differ in one respect that is easy to overlook. The coefficients in Eq. 1 sum to one; those of subtraction sum to zero. Latents are defined only up to a choice of origin, and a shift of that origin cancels in the first case but not in the second. Section III makes this precise for arbitrary nonlinear f and g: recentering the latent space at $f(0)$ leaves the autoencoding function and every unit-sum combination pointwise identical, while mapping subtraction to $g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}) + f(0))$. Any change a subtraction score shows under this map is therefore a property of the coordinate system, not of the model as a function.

We run that intervention on five fixed checkpoints of a waveform GAN autoencoder, spanning two compression rates and three training objectives, with no retraining and no change to the evaluation audio. Our contributions:

1) A reparameterization argument (Section III) showing that subtraction-style latent arithmetic is coordinate-dependent while reconstruction and convex mixing are not, and identifying the model's own reconstruction of the target as the coordinate-invariant reference.
2) The intervention applied to fixed checkpoints (Section V-A): recentering removes the entire aggregate advantage that mixing supervision appears to have on stem subtraction, on every one of the 49 test tracks, while small stem-dependent differences survive.
3) Evidence that the two probes disagree on the same checkpoints (Section V-B): on convex mixing, which the intervention cannot affect, the supervision does deliver a measurable gain at both compression rates, and it also restores level, not only spectral shape.
4) A methodological correction (Section V-C): the widely used decode-vs-decode SI-SDR is inflated by an adversarial term and by decoder sampling noise; we report a ground-truth-referenced variant.

II. Related Work

Neural audio codecs. DAC [3], EnCodec [4] and SoundStream [6] pair convolutional encoders with HiFi-GAN decoders [7] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization also precludes latent interpolation. RAVE [8] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [9] encourage generic smoothness rather than a specific algebraic identity. Mixing-equivariance follows the symmetry-based view of representation learning [10], though convex mixing is an affine combination, not a group action. **Linear audio representations.** Music2Latent [11] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [5] make it approximately linear by conditioning the decoder on the average of separately encoded latents while denoising the true mix under the unchanged consistency loss, a principle related to mixup [12]. Their additivity and separation scores compare two decodes, a protocol Section V-C shows to be confounded, and their separation is negative on every stem (-0.3 to -1.8 dB). No prior work reports latent arithmetic against a declared origin.

III. The Latent Origin Is a Free Coordinate

Proposition. Let $f : \mathcal{X} \to \mathcal{Z}$ and $g : \mathcal{Z} \to \mathcal{X}$ be arbitrary maps, and define the recentered pair

$$f_c(x) = f(x) - f(0), \qquad g_c(z) = g(z + f(0))\,. \tag{2}$$

Then (i) $g_c(f_c(x)) = g(f(x))$ for every x; (ii) for any $x_1, \ldots, x_n$ and any coefficients with $\sum_i \alpha_i = 1$, $g_c(\sum_i \alpha_i f_c(x_i)) = g(\sum_i \alpha_i f(x_i))$; and (iii) for the coefficients $(1, -1)$,

$$g_c(f_c(x) - f_c(y)) = g(f(x) - f(y) + f(0))\,. \tag{3}$$

Proof. $f_c(x) + f(0) = f(x)$ gives (i). For (ii), $\sum_i \alpha_i f_c(x_i) = \sum_i \alpha_i f(x_i) - f(0)\sum_i \alpha_i = \sum_i \alpha_i f(x_i) - f(0)$, and g_c adds $f(0)$ back. For (iii) the coefficients sum to zero, so the offset cancels in the latent difference, $f_c(x) - f_c(y) = f(x) - f(y)$, and g_c adds $f(0)$ inside g. □

Three consequences shape the rest of the paper. First, (f_c, g_c) is *the same autoencoder* as (f, g) as far as reconstruction and Eq. 1 are concerned: by (i) and (ii) those quantities are pointwise identical, for arbitrary nonlinear networks, with no retraining and no approximation. Second, by (iii) subtraction is not: recentering shifts it by one encode of silence, and Section V-A measures a several-decibel shift. A model ranking obtained from raw subtraction can therefore be altered by a change of coordinates that provably leaves reconstruction and mixing untouched, so such a ranking is not by itself a statement about the representation. Third, when f happens to be affine, $f(x) = Ax + b$, we have $f(x) - f(y) + f(0) = f(x - y)$ and Eq. 3 decodes exactly $g(f(x_{\mathrm{res}}))$: the model's own reconstruction of the residual, whatever the decoder does. That quantity, the encode–decode reconstruction of the target, is the reference we report against; the shortfall from it is the *linearity tax*. How far a real encoder departs from affine is empirical, so the recovery is exact only in the affine limit and approximate here.

IV. Experimental Setup

A. Models and supervision

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [7], combined multi-scale-STFT and multi-period discriminator) at 44.1 kHz on 1-second mono chunks, trained on a ∼950-hour union of FMA-large, MAESTRO and MUSDB18-HQ [13].[1] Two compression regimes are used: 7.66× (d=128) and 15.3× (d=64).

Mixing supervision is the explicit-loss form of the additivity principle of [5]. We form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0, 1]$; with $\bar{x} = \alpha x_i + (1-\alpha) x_j$ and $\bar{z} = \alpha f(x_i) + (1-\alpha) f(x_j)$, the decode-mixing loss is, as implemented,

$$\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} = \mathcal{L}_{\mathrm{recon}}(g(\bar{z}), \bar{x}) + \|g(\bar{z}) - \bar{x}\|_1, \tag{4}$$

where $\mathcal{L}_{\mathrm{recon}}$ is the weighted multi-resolution STFT and mel loss also used for reconstruction; the mixed path carries the extra unweighted waveform ℓ_1 term, the ordinary reconstruction path does not. Gradients flow through g and into f via $\bar{z}$. The total objective is $\mathcal{L} = \mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{GAN}} + \lambda \mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} + \lambda_{\ell_2} \|z\|^2$; a variant additionally applies the discriminator to the mixed path (“+disc”). Mixing losses are added either as a 25k-step fine-tune over a converged baseline (7.66×) or from scratch over 250k steps (15.3×), each against a matched control that is identical except that mixing is disabled. All runs use a single NVIDIA RTX 5090.

B. Metrics

We report reconstruction SI-SDR$_{\mathrm{rec}} = \mathrm{SISDR}(g(f(x)), x)$ and, at α=0.5, two equivariance views: SI-SDR$_{\mathrm{lin}} = \mathrm{SISDR}(g(\bar{z}), g(f(\bar{x})))$ (decode-vs-decode, relative to the model's own reconstruction of the mix) and SI-SDR$^{\mathrm{gt}}_{\mathrm{lin}} = \mathrm{SISDR}(g(\bar{z}), \bar{x})$ (vs. ground truth, the quantity in Eq. 1 up to scale and comparable across models). We also report $\ell_{\mathrm{lat}} = \|\bar{z} - f(\bar{x})\|^2 / \|f(\bar{x})\|^2$ and MixRate $= \mathcal{L}_{\mathrm{mix}}(g(\bar{z}), \bar{x}) / \mathcal{L}_{\mathrm{mix}}(g(f(\bar{x})), \bar{x})$, with $\mathcal{L}_{\mathrm{mix}}$ the three-term mixed-path loss of Eq. 4 in both numerator and denominator (< 1 beats the encode-then-decode path). SI-SDR$_{\mathrm{lin}}$ shares the decode-vs-decode structure of prior additivity metrics [5] and can be moved without ground-truth equivariance moving (Section V-B); SI-SDR$^{\mathrm{gt}}_{\mathrm{lin}}$ and MixRate are our metrics of record. SI-SDR fits a gain before scoring, so it does not assess level preservation; we report the fitted gain separately.

C. Protocol and what the intervals cover

Subtraction is evaluated on the MUSDB18 test set, 49 of 50 tracks (one is excluded because its stems sum to the mixture at only 18.5 dB), giving 5,880 (track, stem, chunk) units; chunk selection is a pure function of the seed and the sorted track list, so all models are measured on identical units and every contrast is paired. Confidence intervals come from a paired cluster bootstrap over the 49 tracks (10^4 resamples), the correlated unit. They are a *conditional diagnostic*: they cover evaluation variance over test audio for these checkpoints, and cover neither training-seed variance nor checkpoint selection.

That selection needs disclosure. The released weights are the checkpoint with the lowest validation loss, and the validation loader read the test split. For every 7.66× run this coincides with the final step, so selection did not alter those weights;

[1] Code, training configurations, pretrained weights, every metric reported here, and audio examples: https://github.com/nurdauletakhanov/musicgen

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+4.0	+3.4	5.2
$+f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
7.66× $+\mathcal{L}_{\text{dec}}$	+5.7	+4.9	+7.3	+5.2	+5.8	2.9
$+f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
7.66× $+\mathcal{L}_{\text{dec}}$+disc	+6.2	+5.4	+7.5	+5.2	+6.1	2.6
$+f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
15.3× no mix	+1.2	-1.6	+3.0	+2.2	+1.2	6.6
$+f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
15.3× $+\mathcal{L}_{\text{dec}}$+disc	+5.4	+4.7	+6.9	+4.0	+5.2	2.4
$+f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE I

STEM REMOVAL BY LATENT SUBTRACTION (SI-SDR, DB; MUSDB18 TEST, 5,880 UNITS FROM 49 TRACKS). "$+f(0)$" ROWS READ THE *same* CHECKPOINT IN RECENTERED COORDINATES. "GAP" IS THE LINEARITY TAX AGAINST EACH MODEL'S OWN ENCODE–DECODE RECONSTRUCTION OF THE RESIDUAL (LOWER IS BETTER).

Treatment	raw	recent.	change [95% CI]	tracks
$\mathcal{L}_{\text{mix}}^{\text{dec}}$, 7.66×	+2.33	-0.03	-2.35 [-2.51, -2.20]	49/49
$\mathcal{L}_{\text{mix}}^{\text{dec}}$+disc, 7.66×	+2.64	-0.07	-2.70 [-2.85, -2.54]	49/49
$\mathcal{L}_{\text{mix}}^{\text{dec}}$+disc, 15.3×	+4.02	+0.01	-4.01 [-4.20, -3.82]	49/49

TABLE II

THE INTERVENTION ON FIXED CHECKPOINTS. ADVANTAGE IN STEM-SUBTRACTION SI-SDR (DB) OF EACH MIXING-TRAINED MODEL OVER THE MATCHED CONTROL AT THE SAME COMPRESSION RATE TRAINED WITHOUT MIXING SUPERVISION, BEFORE AND AFTER RECENTERING, WITH THE PAIRED CHANGE AND THE NUMBER OF TRACKS ON WHICH THE ADVANTAGE SHRINKS. RECENTERING IS A CHANGE OF COORDINATES ONLY: NO RETRAINING, SAME UNITS, SAME AUDIO.

both 15.3× runs were selected at step 240k of 250k. Contrasts are between matched configurations under identical selection, and the origin intervention is applied to fixed checkpoints so it is unaffected, but absolute generalization claims should be read with this in mind. The code now supports a held-out validation split.

V. RESULTS

A. Recentering removes the subtraction advantage

We remove a stem by latent subtraction, $\hat{x} = g(f(x_{\text{mix}}) - f(x_{\text{stem}}))$, and score SI-SDR against the true residual. This is an *oracle* setting: the removed stem is available and encoded, so it probes the representation's arithmetic, not blind separation; its use is latent-domain editing when stem codes exist. The intervention of Section III replaces this by $g(f(x_{\text{mix}}) - f(x_{\text{stem}}) + f(0))$, which by Eq. 3 is the same model read in recentered coordinates.

In raw coordinates the supervision looks decisive. Mixing supervision raises subtraction quality from $+3.4$ to $+5.8$ dB at 7.66× and from $+1.2$ to $+5.2$ dB at 15.3× (Table I), advantages of $+2.33$ and $+4.02$ dB on paired units (Table II).

Recentering erases it. Adding $f(0)$ gains $+2.0$ to $+6.0$ dB for every evaluated configuration and leaves each within 0.4–0.5 dB of its own reconstruction reference in aggregate; the models that gain most are the ones that looked worst. The between-model advantage falls to -0.03, -0.07 and $+0.01$ dB, all three intervals containing zero (Table II), and the reduction holds on 49 of 49 tracks in every pair (Fig. 1, right). The no-mixing models also carry the larger offset ($\|f(0)\|$ 6.6 versus 5.4 at 7.66×, 18.1 versus 14.8 at 15.3×), though offset norm alone does not determine the decoded error, which also depends on latent scale and decoder sensitivity.

What survives. Recentering equalizes the aggregate, not everything. Per stem the recentered contrast is still nonzero in both directions: at 15.3×, $+0.35$ dB on bass and -0.34 dB on vocals; at 7.66×, $+0.37$ on "other" against -0.31 on drums. Mixing supervision moves where the arithmetic error sits across sources without moving its mean. The residual 0.4–0.5 dB gap to the reference is what decoding leaves after recentering, not a direct measure of encoder non-affinity; per track the tail is wider (worst 2.2 dB, 29–32 of 49 tracks within 0.5 dB).

Baselines without latent arithmetic. The unchanged mixture scores $+17.1$ dB against the residual, above every model's reconstruction reference, because the removed stem is a small share of the energy: the metric certifies arithmetic only relative to the model's own reconstruction. Autoencoding the mixture unchanged scores $+2.2$ dB ($+1.7$ at 15.3×), decoding both signals and subtracting waveforms $+7.5$ ($+6.5$). Recentered latent subtraction beats waveform subtraction by $+0.7$ to $+0.9$ dB (intervals exclude zero), raw subtraction loses to it everywhere.

B. Convex mixing tells a different story

By part (ii) of the proposition, recentering cannot change any quantity in Eq. 1. The mixing probe therefore reports on the autoencoder itself, and there the supervision does something (Table III). At 7.66×, $\mathcal{L}_{\text{mix}}^{\text{dec}}$ improves ground-truth equivariance from 8.0 to 10.2 dB and is the only configuration with MixRate < 1; reconstruction SI-SDR spans 11.2–11.4 dB and FAD 0.044–0.046, so we observe no reconstruction cost at this rate, on single runs that cannot establish the absence of a difference this small. Trained from scratch at 15.3× against a matched control, the combined $\mathcal{L}_{\text{mix}}^{\text{dec}}$+disc recipe improves SI-SDR$_{\text{lin}}^{\text{gt}}$ from 6.8 to 8.6 dB, MixRate from 1.19 to 1.05 and encoder linearity ℓ_{lat} from 0.078 to 0.057 at equal reconstruction (9.9 vs. 10.0 dB), while FAD rises from 0.052 to 0.061; whether that is perceptible is untested. The 15.3× pair differs by the whole combined recipe, so it attributes the gain to that recipe against its control and does not isolate the contribution of $\mathcal{L}_{\text{mix}}^{\text{dec}}$ alone at that rate.

Level, not only shape. SI-SDR discards gain, so we also report the gain it fits: the decoded mix is 1.6 dB too quiet without the loss and 0.9 dB with it, against the model's own reconstruction level error of 0.8 dB; at 15.3×, 1.8 versus 1.0 against 0.9.

The mixed-path discriminator buys the metric, not the property. At 7.66×, adding an adversarial term on the mixed path leaves ground-truth equivariance unchanged (10.2 → 9.9 dB, MixRate 0.97 → 1.05) while raising decode-vs-decode SI-SDR$_{\text{lin}}$ from 12.1 to 16.0 dB. The two decode paths move

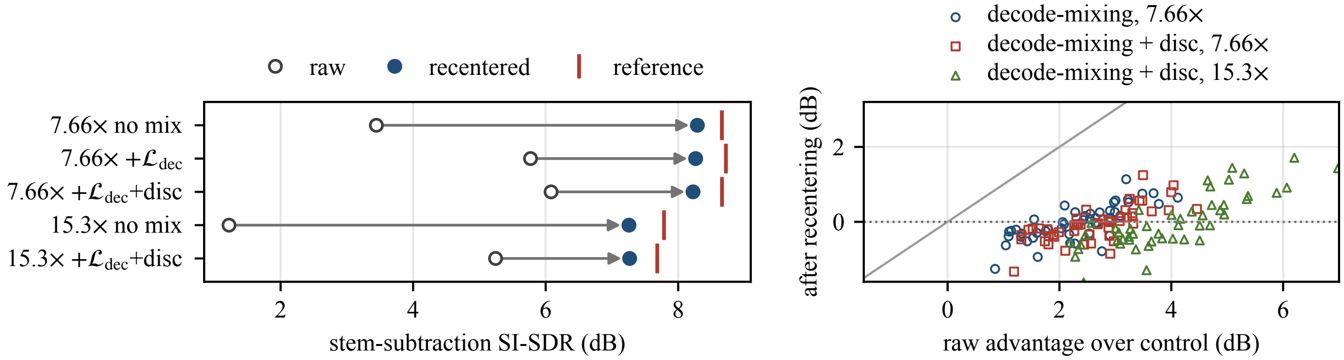

Fig. 1. The origin intervention on five fixed checkpoints. *Left:* stem-subtraction SI-SDR moves from raw coordinates to recentered ones (arrow), landing at each model's own reconstruction of the residual (red tick); the raw ordering does not survive. *Right:* per-track advantage over the matched no-mixing control, raw (horizontal) against recentered (vertical), one point per track. Every point falls below the diagonal and the recentered values sit at zero.

Model	SDR_{rec}	SDR_{lin}	$\text{SDR}_{\text{lin}}^{\text{gt}}$	MixRate	FAD
GAN, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN, 7.66×, +$\mathcal{L}_{\text{dec}}$	+11.4	+12.1	+10.2	0.972	0.044
GAN, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN, 15.3×, +$\mathcal{L}_{\text{dec}}$+disc	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	0.134
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE III
CONVEX MIXING ACROSS COMPRESSION AND ARCHITECTURE (FULL TEST SET, α=0.5, EMA WEIGHTS FOR MUSIC2LATENT). THE M2L "FINE-TUNE CONTROL" IS THE CONTINUED-TRAINING BASELINE WITH MIXING DISABLED. THESE QUANTITIES ARE INVARIANT TO THE INTERVENTION OF SECTION III.

toward each other without either moving toward $\bar{x}$, so the two metrics rank these configurations differently. We do not identify the mechanism. Whether the discriminator has an effect without $\mathcal{L}_{\text{mix}}^{\text{dec}}$ is not tested here.

Architecture. Fine-tuning the published Music2Latent [11] consistency autoencoder with the decode-mixing objective improves $\text{SI-SDR}_{\text{lin}}^{\text{gt}}$ against a continued-training control ($-9.5 \rightarrow -7.8$ dB) but stays far from usable waveform arithmetic; its subtraction is unusable outright (-11.2 dB, Table I). The systems differ in encoder, loss family, compression and training data as well as decoder class, so the contrast isolates none of them: we report it as a limit on transfer, not an explanation.

C. Decode-vs-decode scores are protocol-sensitive

For stochastic decoders the decode-vs-decode metric is confounded a second way. A consistency-model decoder [11] draws fresh noise per call: decoding the *same* latent twice is deterministic under shared noise but scores -1.8 dB under independent noise, with no latent arithmetic at all. On identical pairs (Fig. 2) the same checkpoints score -9.1 and -7.3 dB when the two decodes draw independent noise and $+3.6$ and $+5.9$ dB when they share it: a 13 dB swing driven by the evaluation protocol rather than the model. Which protocol prior work used is not stated in enough detail for us to determine. We use shared decode noise everywhere.

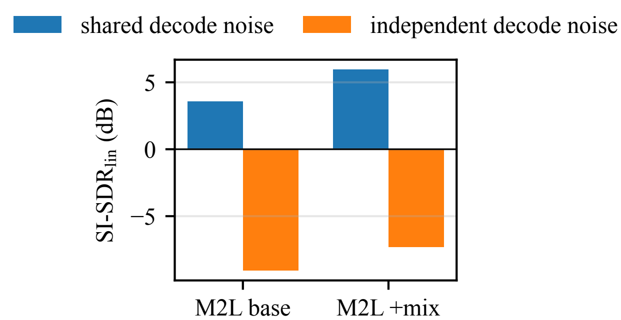

Fig. 2. Decode-noise protocol confound (Music2Latent, MUSDB18, α=0.5). The same checkpoints score deeply negative under independent-noise decoding and positive under shared-noise decoding; the negative scores reflect decoder sampling noise rather than latent linearity.

VI. CONCLUSION

Latent stem subtraction is measured in coordinates that the model never fixes. Recentering the latent space at the encoded silence leaves the autoencoding function and every unit-sum combination pointwise unchanged, for arbitrary nonlinear encoders and decoders, yet it moves subtraction scores by several decibels and, on five fixed checkpoints, removes the entire aggregate advantage that mixing supervision appeared to have there, on all 49 test tracks. What the supervision buys is convex equivariance, which the intervention cannot affect and which improves at both rates. We therefore recommend reporting latent arithmetic against a declared origin and against the model's own reconstruction of the target. **Limitations.** Comparisons are single-run at one initialization; checkpoint selection used the test split (Section IV-C); intervals are conditional on these checkpoints; recovery is exact only for an affine encoder; audio is mono and untested by listeners.

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